



3.6 Core Logic

```
def generate_account():
    while True:
        num = math.floor(random.random() * (10 ** 10))
        account_no = "A" + str(num)
        if account_no not in ACCOUNTS:
            return account_no # ensures every account number is unique

def withdraw(account, amount):
    if amount <= USERS[account]['balance']:
        USERS[account]['balance'] -= amount
        return True # withdrawal allowed, balance updated
    return False # insufficient funds, transaction rejected
```

The generate_account() function creates a unique account number by generating a random 10-digit value and ensuring it does not already exist. The withdraw() function allows transactions only when sufficient balance is available, preventing negative account balances.

3.7 Output Screenshot